

6(a)	<i>mark first two statements given</i> cellulose, cell wall / lining, allows adhesion of water ; thick (cellulose) cell wall prevents collapse / <i>idea of</i> providing support (under tension) ; (cell wall impregnated with) lignin, for waterproofing / prevents water loss ; lignin, rings / spirals / thickening / AW (of walls), prevents collapse / <i>idea of</i> providing support (under tension) ; A lignified walls no cytoplasm / lack of contents / hollow / empty (lumen), gives, less resistance to / unimpeded / uninterrupted / unhindered / ease of, flow ; A greater volume per unit time / faster rate R continuous, smooth R dead lack of end walls / continuous tube, so, less resistance to / unimpeded / uninterrupted / unhindered / ease of, flow ; AW R continuous, smooth pits / pores, for lateral movement / for movement around air bubbles / supplies (water) to (surrounding), cells / tissues ; R holes wide / large diameter / large lumen, so large volume of water can be transported ; <i>if max 2 not gained, allow one mark for two correct explanations or descriptions where accompanying descriptions and explanations are in context</i>	2
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Question	Answer	Marks
6(b)	<i>two from</i> <i>accept phloem vessels for sieve tubes</i> only sieve tubes / one type, v, arteries, veins and capillaries (any two) A three types; sieve tubes are composed of cells v blood vessels composed of tissues ; A ref. to named tissues sieve tubes, have cytoplasm or blood vessels are, hollow / AW; sieve tubes have sieve plates ; ora sieve tubes have companion cells (to fully function) ; veins have valves ; ora <i>ref. to</i> blood vessels and vasoconstriction ; ora <i>idea of transport</i> open (phloem) v closed (blood vessels) ; A sieve tubes have plasmodesmata <i>ref. to</i> unidirectional flow in blood vessels ; ora	2
6(c)	<i>three from</i> mitochondria, synthesise ATP / carry out aerobic respiration ; more ATP needs to be synthesised / increased ATP synthesis ; A need to produce large quantities of ATP to provide more (metabolic) energy ; R <i>in context of water uptake</i> for active transport / active uptake ; <i>in context of mineral ions</i> A to move mineral ions against the concentration gradient	3